

Assignment No 09

AIM: Create a Java application that connects to a MySQL/PostgreSQL database to manage employee information. Implement functionalities like adding, updating, deleting, and viewing employee records using JDBC. Use prepared statements to prevent SQL injection and handle exceptions gracefully.

Code:

[illegible]

```
        rs.getString("dept_name")
    );
}
break;
```

case 2:

```
System.out.print("Enter ID: ");
int id = sc.nextInt();
sc.nextLine();
```

```
System.out.print("Enter Name: ");
String name = sc.nextLine();
```

```
System.out.print("Enter Salary: ");
double salary = sc.nextDouble();
```

```
System.out.print("Enter Dept ID: ");
int deptId = sc.nextInt();
```

```
String insert = "INSERT INTO emp VALUES (" + id + ", '" + name + "', " +
salary + ", " + deptId + ")";
conn.s.executeUpdate(insert);
```

```
System.out.println("Employee Inserted!");
break;
```

case 3:

```
System.out.print("Enter ID to update: ");
int uid = sc.nextInt();
```

```
System.out.print("Enter New Salary: ");
double newSalary = sc.nextDouble();
```

```
String update = "UPDATE emp SET salary=" + newSalary + " WHERE
emp_id=" + uid;
conn.s.executeUpdate(update);
```

```
System.out.println("Salary Updated!");
break;
```

case 4:

```
System.out.print("Enter ID to delete: ");
int did = sc.nextInt();
```

```
String delete = "DELETE FROM emp WHERE emp_id=" + did;
conn.s.executeUpdate(delete);
```

```

        System.out.println("Employee Deleted!");
        break;

    case 5:
        System.out.print("Enter Start ID: ");
        int start = sc.nextInt();

        System.out.print("Enter End ID: ");
        int end = sc.nextInt();

        ResultSet rs1 = conn.s.executeQuery(
            "SELECT e.emp_id, e.name, e.salary, d.dept_name " +
            "FROM emp e JOIN dept d ON e.dept_id = d.dept_id " +
            "WHERE e.emp_id BETWEEN " + start + " AND " + end
        );

        System.out.println("\nID   Name   Salary   Department");

        while (rs1.next()) {
            System.out.println(
                rs1.getInt("emp_id") + " " +
                rs1.getString("name") + " " +
                rs1.getDouble("salary") + " " +
                rs1.getString("dept_name")
            );
        }
        break;

    case 6:
        System.out.println("Exiting...");
        break;

    default:
        System.out.println("Invalid choice!");
    }

} while (choice != 6);

conn.c.close();

} catch (Exception e) {
    e.printStackTrace();
}
}
}

```

Output:

```
===== EMPLOYEE MENU =====
1. Display All Employees
2. Insert Employee
3. Update Employee Salary
4. Delete Employee
5. Search Employees (ID Range)
6. Exit
Enter choice: 1
```

```
ID  Name  Salary  Department
101 Amit  45000.0 HR
102 Sneha 60000.0 IT
103 Rahul 55000.0 IT
104 Priya 0.0 Finance
106 Neha 52000.0 Marketing
107 Rohan 48000.0 Sales
108 Anjali 70000.0 IT
109 Vikas 39000.0 Admin
110 Meena 62000.0 Operations
111 Suresh 58000.0 R&D
112 Pooja 0.0 IT
113 Arjun 75000.0 Logistics
114 Divya 68000.0 Security
115 Nikhil 47000.0 Legal
```

```
===== EMPLOYEE MENU =====
1. Display All Employees
2. Insert Employee
3. Update Employee Salary
4. Delete Employee
5. Search Employees (ID Range)
6. Exit
Enter choice: 2
Enter ID: 123
Enter Name: Hire 0m
Enter Salary: 25000
Enter Dept ID: 2
Employee Inserted!
```

```
===== EMPLOYEE MENU =====  
1. Display All Employees  
2. Insert Employee  
3. Update Employee Salary  
4. Delete Employee  
5. Search Employees (ID Range)  
6. Exit  
Enter choice: 3  
Enter ID to update: 120  
Enter New Salary: 1300000  
Salary Updated!
```

```
===== EMPLOYEE MENU =====  
1. Display All Employees  
2. Insert Employee  
3. Update Employee Salary  
4. Delete Employee  
5. Search Employees (ID Range)  
6. Exit  
Enter choice: 4  
Enter ID to delete: 2  
Employee Deleted!
```

```
===== EMPLOYEE MENU =====
1. Display All Employees
2. Insert Employee
3. Update Employee Salary
4. Delete Employee
5. Search Employees (ID Range)
6. Exit
Enter choice: 5
Enter Start ID: 101
Enter End ID: 107
```

ID	Name	Salary	Department
101	Amit	45000.0	HR
102	Sneha	60000.0	IT
103	Rahul	55000.0	IT
104	Priya	0.0	Finance
106	Neha	52000.0	Marketing
107	Rohan	48000.0	Sales

```
===== EMPLOYEE MENU =====
1. Display All Employees
2. Insert Employee
3. Update Employee Salary
4. Delete Employee
5. Search Employees (ID Range)
6. Exit
Enter choice: 6
Exiting...

Process finished with exit code 0
```